

1. Sample Details

The malware sample analyzed in this report is ArdamaxKeylogger.exe, a known commercial keylogging software often misused for malicious purposes such as credential theft and user surveillance. Source of Sample

<https://github.com/ytisf/theZoo/tree/master/malware/Binaries/Keylogger.Ardamax>

File Information

- File Name: ArdamaxKeylogger.exe
- Type: Executable (.exe)
- Category: Keylogger
- Behavior: Records keystrokes and stores data locally

2. Tools Used

The following tools were used during the analysis: Static Analysis Tools

- Hash generation tools
- Detect It Easy (DIE)
- CFF
- Ghidra

Dynamic Analysis Tools

- Process Monitor (ProcMon) – for tracking file system, registry, and process activity
- Command Prompt (cmd) – for registry inspection
- x32 dbg
- Fake Net
- Wireshark

Environment

- FLARE VM (Windows-based malware analysis environment) ☐ Kali

3. Statics Analysis

- The malware sample Keylogger_Ardamax.zip was verified using MD5 and SHA256 hashing.
- Hash values were generated using md5sum and sha256sum commands.
- The computed hashes matched exactly with the provided .md5 and .sha256 files.
- This confirms that the sample was not tampered with or corrupted.

- Hash verification ensures the integrity and authenticity of the file.
- It guarantees that further analysis is performed on a trusted and original sample.

```

st@kali: ~/Downloads
File Actions Edit View Help
(st@kali)~[~]
$ cd Downloads
(st@kali)~[~/Downloads]
$ ls
Keylogger.Ardamax.md5  Keylogger.Ardamax.pass  Keylogger.Ardamax.sha256  Keylogger.Ardamax.zip  keylogger.exe
(st@kali)~[~/Downloads]
$ cat Keylogger.Ardamax.md5
5de75a478ffb3aa01a88f4e539f3edc0  Keylogger.Ardamax.zip
(st@kali)~[~/Downloads]
$ md5sum Keylogger.Ardamax.zip
5de75a478ffb3aa01a88f4e539f3edc0  Keylogger.Ardamax.zip
(st@kali)~[~/Downloads]
$ cat Keylogger.Ardamax.sha256
9c662e2c950e9cba8367a47f628553291fie26b7e897a8533c00a4b27e174227  Keylogger.Ardamax.zip
(st@kali)~[~/Downloads]
$ sha256sum Keylogger.Ardamax.zip
9c662e2c950e9cba8367a47f628553291fie26b7e897a8533c00a4b27e174227  Keylogger.Ardamax.zip
(st@kali)~[~/Downloads]
$

```

- Strings analysis was performed on the malware sample to identify embedded readable content.
- The output contained mostly random and unreadable characters, indicating possible obfuscation or packing.
- However, some readable API functions such as LoadLibrary, ReadFile, WriteFile, and CreateFile were identified.

```

File Actions Edit View Help
SrXF
1Rich
.text
.rdata
@.data
.rsrc
SVWh
SrXFt
X^[
~$+~8YY
W9^4t
W$YY
G0_[
W$YY3

```

```

LoadLibraryW
ReadFile
WriteFile
DeleteFileW
FreeLibrary
GetTempFileNameW
CloseHandle
SetFilePointer
CreateFileW
GetModuleFileNameW
GetModuleHandleW
GetTempPathW
GetModuleHandleA
GetStartupInfoA
KERNEL32.dll
MessageBoxW
USER32.dll
.?AVI Loader@@
.?AVC Loader@@
.?AV?$CStreamDecompress@VCWin32FileReader@@VCWin32FileWriter@@VCStreamSimpleCallback@@@@

```

- Presence of system DLLs like KERNEL32.dll and USER32.dll suggests interaction with Windows system functions.
- The mixture of readable and unreadable strings indicates that the malware is likely packed or obfuscated.
- This technique is commonly used by malware to hide its actual functionality and evade detection.

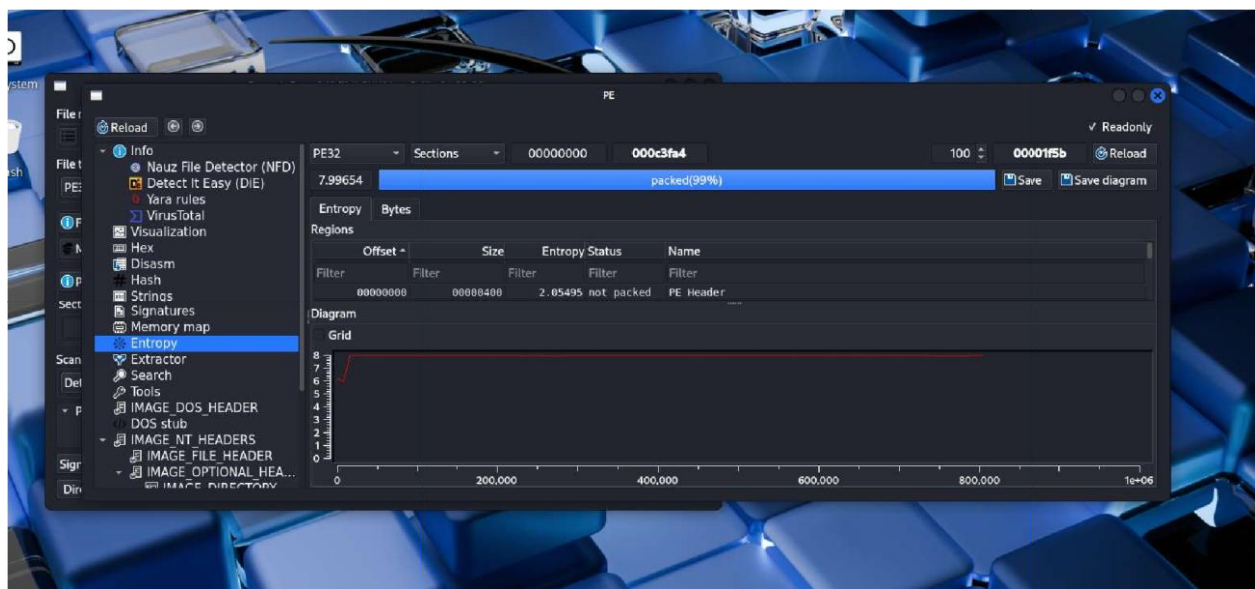
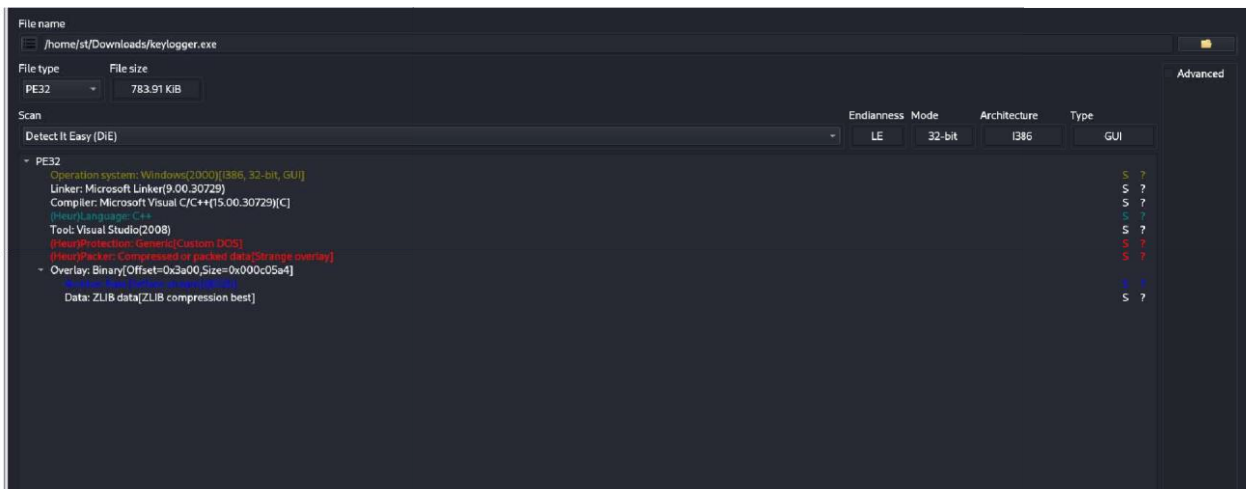
```

File Actions Edit View Help
ze>`
Y+L+
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+XJo
(xDB
/>{,
+*pj
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?i@T
owB1
E3"$
TwS%H
`yUODw:
'DH'
>J8k716
-0C+
Jg`)"
jFC>()
5~A^
Ff61
[wRu
ShmOuM
HfE.
jR]
az7S
%5>H{
zs2f
yx~l
x7F
S9(6
we6{Q
oJDL
SH#V
R6`?
km(o
Ws{SAvm
FocI
q{:$
t:NpD
5WpVW
!d5e
:|

```

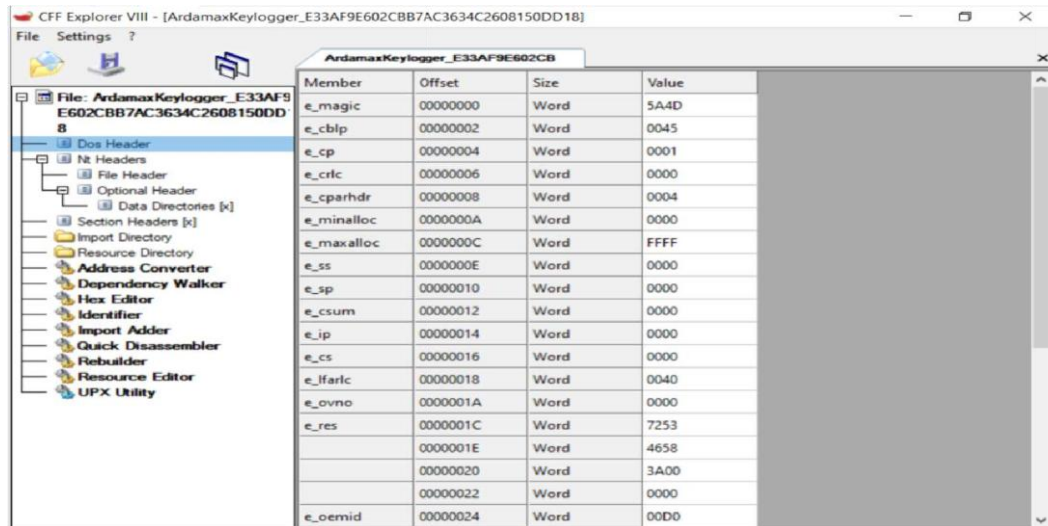
IDE

- The file is identified as a PE32 (32-bit) executable using detect it easy (DIE) analysis tool. ☐ Detect It Easy flagged the file as compressed/packed with a suspicious overlay.
- Entropy analysis shows a very high value (~7.99), indicating strong packing or encryption.
- The presence of ZLIB-compressed data and a non-standard overlay suggests hidden content embedded within the file.
- This confirms the malware uses obfuscation techniques to evade detection and hinder analysis.



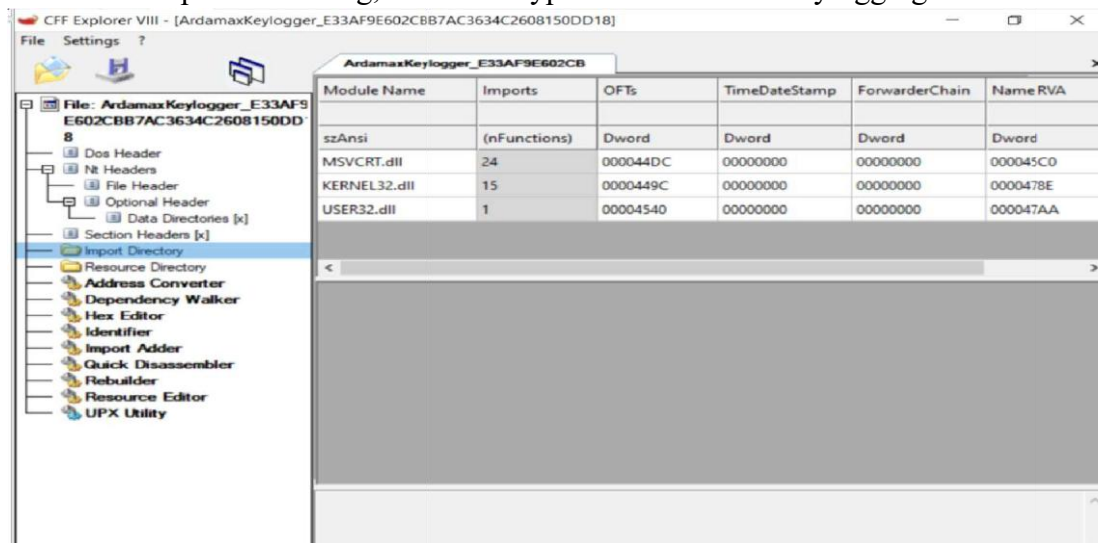
CFF

- The executable contains a valid DOS header (MZ signature – 5A4D), confirming it is a legitimate Windows PE file.



Member	Offset	Size	Value
e_magic	00000000	Word	5A4D
e_cblp	00000002	Word	0045
e_cp	00000004	Word	0001
e_crlc	00000006	Word	0000
e_cparhdr	00000008	Word	0004
e_minalloc	0000000A	Word	0000
e_maxalloc	0000000C	Word	FFFF
e_ss	0000000E	Word	0000
e_sp	00000010	Word	0000
e_csum	00000012	Word	0000
e_ip	00000014	Word	0000
e_cs	00000016	Word	0000
e_lfarlc	00000018	Word	0040
e_ovno	0000001A	Word	0000
e_res	0000001C	Word	7253
	0000001E	Word	4658
	00000020	Word	3A00
	00000022	Word	0000
e_oemid	00000024	Word	00D0

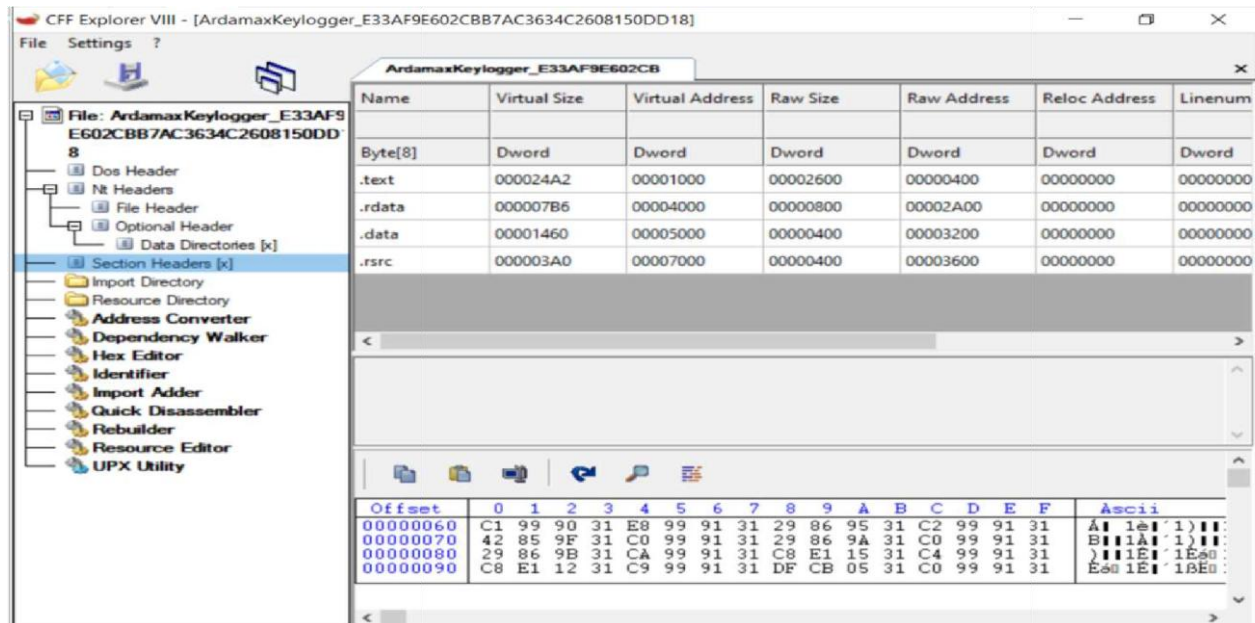
- The executable imports key Windows libraries such as KERNEL32.dll, USER32.dll, and MSVCRT.dll, indicating interaction with system-level functions.
- These imports suggest capabilities like file handling, memory management, and user input monitoring, which are typical behaviors of keylogging malware.



Module Name	Imports	OFs	TimeStamp	ForwarderChain	Name RVA
szAnsi	(nFunctions)	Dword	Dword	Dword	Dword
MSVCRT.dll	24	000044DC	00000000	00000000	000045C0
KERNEL32.dll	15	0000449C	00000000	00000000	0000478E
USER32.dll	1	00004540	00000000	00000000	000047AA

- Uses USER32.dll → user input
- Uses KERNEL32.dll → file operations
- Packed → hiding real behavior

- **SECTION HEADER :-** The executable contains standard sections like .text, .rdata, .data, and .rsrc, which represent code, read-only data, variables, and resources respectively.
- The unusual size distribution and relatively small raw sizes suggest that the actual payload may be compressed or packed, indicating hidden or obfuscated code.



- **OPTIONAL HEADER:-**The binary is identified as a PE32 Windows GUI executable, with the entry point located in the .text section, indicating where execution begins.
- Standard values like ImageBase, SectionAlignment, and Subsystem (Windows GUI) appear normal, suggesting the malware is designed to run as a typical user-level application without raising suspicion.

CFF Explorer VIII - [ArdamaxKeylogger_E33AF9E602CBB7AC3634C2608150DD18]

File Settings ?

ArdamaxKeylogger_E33AF9E602CB

Member	Offset	Size	Value	Meaning
Magic	000000D8	Word	010B	PE32
MajorLinkerVersion	000000DA	Byte	09	
MinorLinkerVersion	000000DB	Byte	00	
SizeOfCode	000000DC	Dword	00002600	
SizeOfInitializedData	000000E0	Dword	00002200	
SizeOfUninitializedData	000000E4	Dword	00000000	
AddressOfEntryPoint	000000E8	Dword	000032EE	.text
BaseOfCode	000000EC	Dword	00001000	
BaseOfData	000000F0	Dword	00004000	
ImageBase	000000F4	Dword	00400000	
SectionAlignment	000000F8	Dword	00001000	
FileAlignment	000000FC	Dword	00000200	
MajorOperatingSystemVersion	00000100	Word	0005	
MinorOperatingSystemVersion	00000102	Word	0000	
MajorImageVersion	00000104	Word	0000	
MinorImageVersion	00000106	Word	0000	
MajorSubsystemVersion	00000108	Word	0005	
MinorSubsystemVersion	0000010A	Word	0000	
Win32VersionValue	0000010C	Dword	00000000	

CFF Explorer VIII - [ArdamaxKeylogger_E33AF9E602CBB7AC3634C2608150DD18]

File Settings ?

ArdamaxKeylogger_E33AF9E602CB

Member	Offset	Size	Value	Meaning
FileAlignment	000000FC	Dword	00000200	
MajorOperatingSystemVersion	00000100	Word	0005	
MinorOperatingSystemVersion	00000102	Word	0000	
MajorImageVersion	00000104	Word	0000	
MinorImageVersion	00000106	Word	0000	
MajorSubsystemVersion	00000108	Word	0005	
MinorSubsystemVersion	0000010A	Word	0000	
Win32VersionValue	0000010C	Dword	00000000	
SizeOfImage	00000110	Dword	00008000	
SizeOfHeaders	00000114	Dword	00000400	
Checksum	00000118	Dword	00000000	
Subsystem	0000011C	Word	0002	Windows GUI
DllCharacteristics	0000011E	Word	8000	Click here
SizeOfStackReserve	00000120	Dword	00100000	
SizeOfStackCommit	00000124	Dword	00001000	
SizeOfHeapReserve	00000128	Dword	00100000	
SizeOfHeapCommit	0000012C	Dword	00001000	
LoaderFlags	00000130	Dword	00000000	
NumberOfRvaAndSizes	00000134	Dword	00000010	

- **DATA DIRECTORIES:-** The presence of a valid Import Directory (.rdata) indicates that the executable relies on external Windows APIs, which may be used for system interaction such as file handling or user input capture.
- Many directories like Debug, TLS, and Security are empty (0 values), which is common in packed or malicious binaries to reduce metadata and hinder detailed analysis.

CFF Explorer VIII - [ArdamaxKeylogger_E33AF9E602CBB7AC3634C2608150DD18]

File Settings ?

File: ArdamaxKeylogger_E33AF9E602CBB7AC3634C2608150DD18

- Dos Header
- Nt Headers
- File Header
- Optional Header
 - Data Directories [x]
 - Section Headers [x]
- Import Directory
- Resource Directory
- Address Converter
- Dependency Walker
- Hex Editor
- Identifier
- Import Adder
- Quick Disassembler
- Rebuilder
- Resource Editor
- UPX Utility

ArdamaxKeylogger_E33AF9E602CB

Member	Offset	Size	Value	Section
Export Directory RVA	00000138	Dword	00000000	
Export Directory Size	0000013C	Dword	00000000	
Import Directory RVA	00000140	Dword	0000444C	.rdata
Import Directory Size	00000144	Dword	00000050	
Resource Directory RVA	00000148	Dword	00007000	.rsrc
Resource Directory Size	0000014C	Dword	000003A0	
Exception Directory RVA	00000150	Dword	00000000	
Exception Directory Size	00000154	Dword	00000000	
Security Directory RVA	00000158	Dword	00000000	
Security Directory Size	0000015C	Dword	00000000	
Relocation Directory RVA	00000160	Dword	00000000	
Relocation Directory Size	00000164	Dword	00000000	
Debug Directory RVA	00000168	Dword	00000000	
Debug Directory Size	0000016C	Dword	00000000	
Architecture Directory RVA	00000170	Dword	00000000	
Architecture Directory Size	00000174	Dword	00000000	
Reserved	00000178	Dword	00000000	
Reserved	0000017C	Dword	00000000	
TLS Directory RVA	00000180	Dword	00000000	

CFF Explorer VIII - [ArdamaxKeylogger_E33AF9E602CBB7AC3634C2608150DD18]

File Settings ?

File: ArdamaxKeylogger_E33AF9E602CBB7AC3634C2608150DD18

- Dos Header
- Nt Headers
- File Header
- Optional Header
 - Data Directories [x]
 - Section Headers [x]
- Import Directory
- Resource Directory
- Address Converter
- Dependency Walker
- Hex Editor
- Identifier
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- UPX Utility

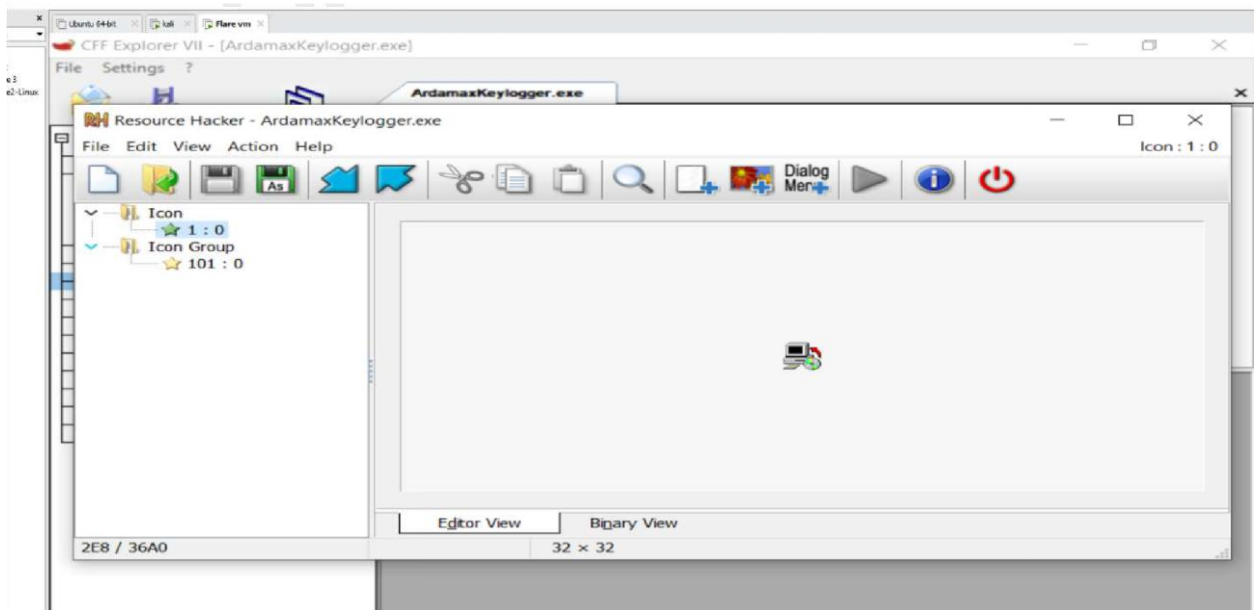
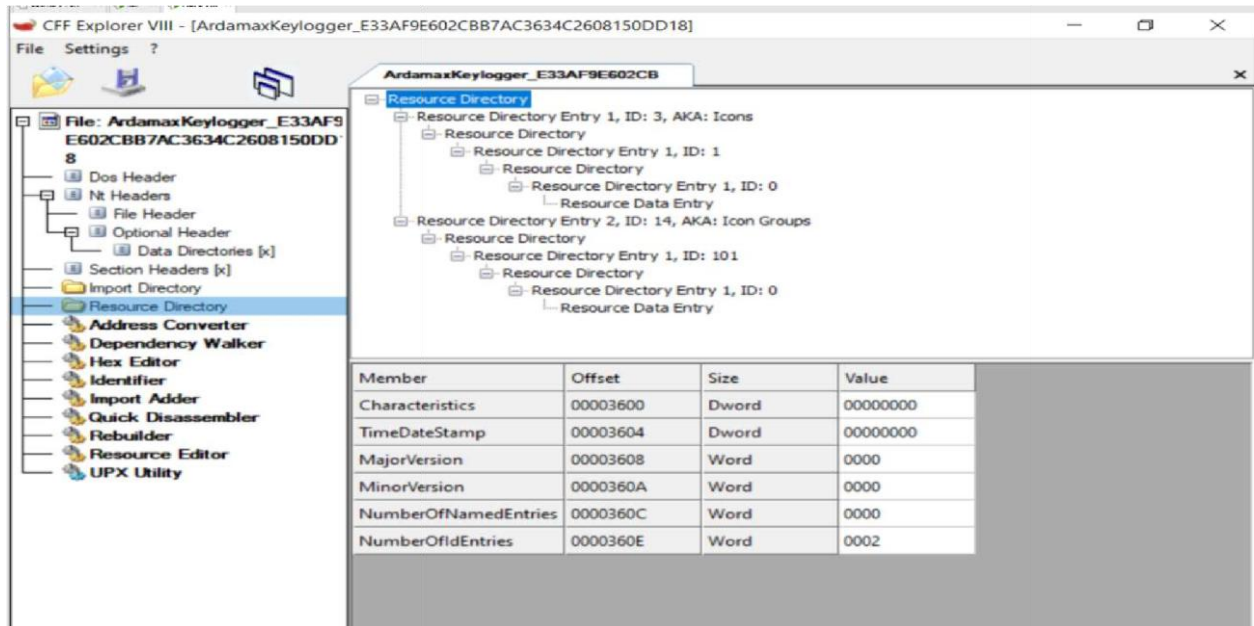
ArdamaxKeylogger_E33AF9E602CB

Member	Offset	Size	Value	Section
Relocation Directory Size	00000164	Dword	00000000	
Debug Directory RVA	00000168	Dword	00000000	
Debug Directory Size	0000016C	Dword	00000000	
Architecture Directory RVA	00000170	Dword	00000000	
Architecture Directory Size	00000174	Dword	00000000	
Reserved	00000178	Dword	00000000	
Reserved	0000017C	Dword	00000000	
TLS Directory RVA	00000180	Dword	00000000	
TLS Directory Size	00000184	Dword	00000000	
Configuration Directory RVA	00000188	Dword	00000000	
Configuration Directory Size	0000018C	Dword	00000000	
Bound Import Directory RVA	00000190	Dword	00000000	
Bound Import Directory Size	00000194	Dword	00000000	
Import Address Table Directory ...	00000198	Dword	00004000	.rdata
Import Address Table Directory ...	0000019C	Dword	000000AC	
Delay Import Directory RVA	000001A0	Dword	00000000	
Delay Import Directory Size	000001A4	Dword	00000000	
.NET MetaData Directory RVA	000001A8	Dword	00000000	
.NET MetaData Directory Size	000001AC	Dword	00000000	

Start

RESOURCE DIRECTORY :-The resource directory contains entries such as Icons and Icon Groups, confirming that the executable includes embedded GUI-related resources.

- The icon appears generic and minimal, which is typical for malware samples that avoid distinctive visual identifiers to reduce suspicion.



Ghidra

- The presence of Windows API functions such as CreateFileW, WriteFile, and DeleteFileW indicates that the executable performs file operations on the system.

□

- These functions suggest that the malware is capable of creating files, storing captured data (such as keystrokes), and deleting traces, which is consistent with keylogger behavior

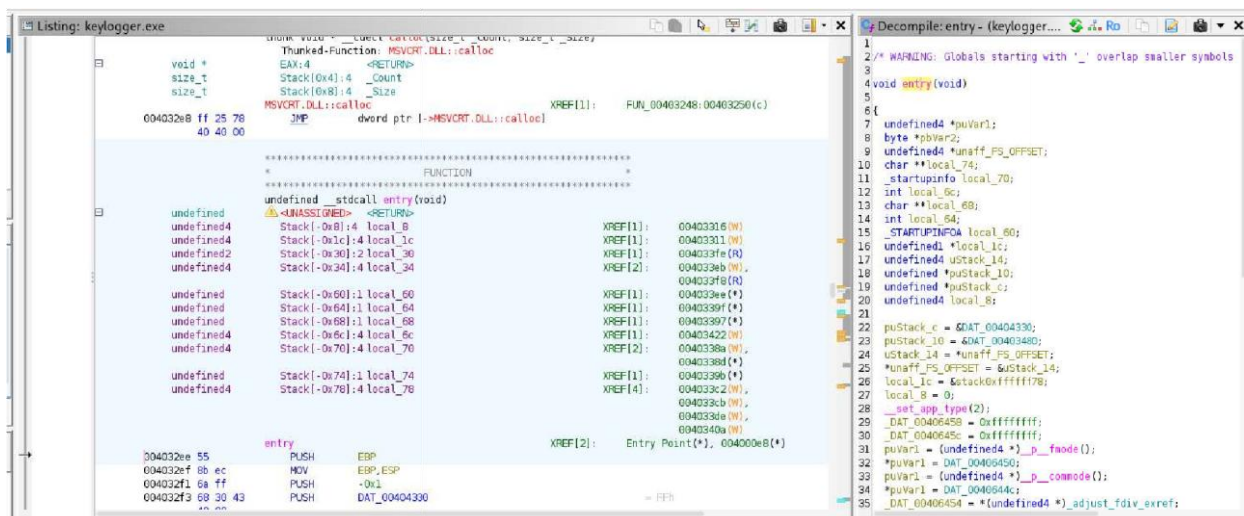
```
*****
HANDLE __stdcall CreateFileW(LPCWSTR lpFileName, DWORD d...
EAX:4 <RETURN>
Stack[0x4]:4 lpFileName
Stack[0x8]:4 dwDesiredAccess
Stack[0xc]:4 dwShareMode
Stack[0x10]:4 lpSecurityAttributes
Stack[0x14]:4 dwCreationDisposition
Stack[0x18]:4 dwFlagsAndAttributes
Stack[0x1c]:4 hTemplateFile
127 CreateFileW <<not bound>>
PTR_CreateFileW_00404010 XREF[1]: FUN_004010be:004010e8(R)
00404010 20 47 00 00 addr KERNEL32.DLL::CreateFileW
*****

*****
* POINTER to EXTERNAL FUNCTION *
*****
BOOL __stdcall DeleteFileW(LPCWSTR lpFileName)
EAX:4 <RETURN>
Stack[0x4]:4 lpFileName
195 DeleteFileW <<not bound>>
PTR_DeleteFileW_00404024 XREF[1]: FUN_0040106a:00401088(R)
00404024 d0 46 00 00 addr KERNEL32.DLL::DeleteFileW

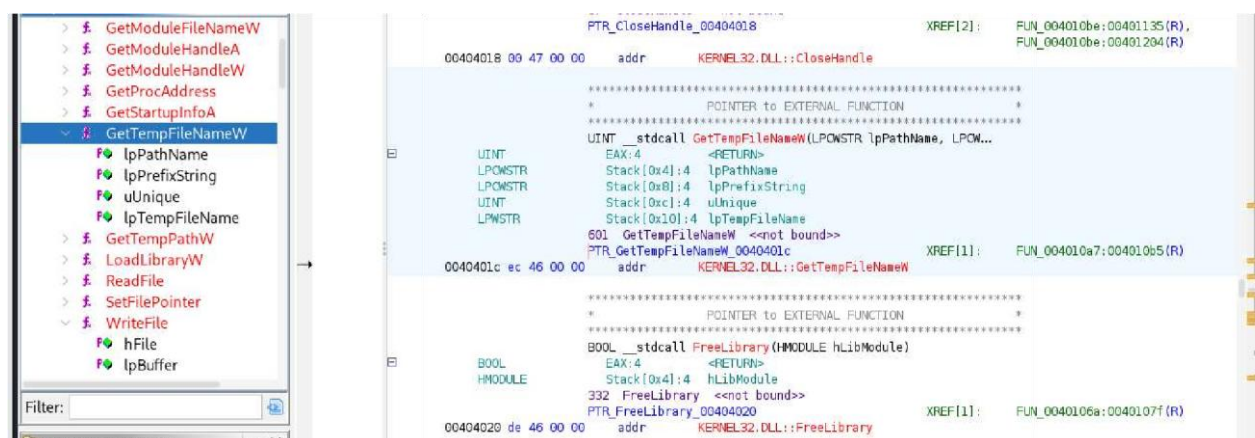
*****
* POINTER to EXTERNAL FUNCTION *
*****
BOOL __stdcall WriteFile(HANDLE hFile, LPCVOID lpBuffer,...
EAX:4 <RETURN>
Stack[0x4]:4 hFile
Stack[0x8]:4 lpBuffer
Stack[0xc]:4 nNumberOfBytesToWrite
Stack[0x10]:4 lpNumberOfBytesWritten
Stack[0x14]:4 lpOverlapped
1165 WriteFile <<not bound>>
PTR_WriteFile_00404028 XREF[1]: FUN_00401047:00401057(R)
00404028 c4 46 00 00 addr KERNEL32.DLL::WriteFile
```

The entry point function shows multiple undefined variables and generic stack allocations, indicating that the code is obfuscated or packed, making analysis difficult.

- The absence of meaningful function names and structured logic suggests that the malware uses anti-analysis techniques to hide its actual behavior from reverse engineers.



- Additional functions like `GetTempPathW` and `GetTempFileNameW` suggest that the malware may create temporary files to store captured data, which is a common technique used by keyloggers.



- `CreateFileW` → create log file
- `WriteFile` → store keystrokes
- `ReadFile` → access stored data
- `DeleteFileW` → remove evidence
- `GetTempPathW` → hide logs in temp folder

4. Dynamic Analysis

ProcMon

- The process `ArdamaxKeylogger.exe` creates a new thread, indicating execution of additional hidden or background functionality.
- It accesses `HKLM\SOFTWARE\Microsoft\Input\Settings`, suggesting interaction with keyboard/input-related configurations, which aligns with keylogging behavior.

Time	Process Name	Operation	Path	Result	Detail
12:50:42	ctfmon.exe	RegQueryKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Query: Handle Tag...
12:50:42	ctfmon.exe	RegOpenKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Desired Access: Q...
12:50:42	ctfmon.exe	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
12:50:42	ctfmon.exe	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
12:50:42	ctfmon.exe	RegQueryKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Query: Handle Tag...
12:50:42	Explorer.EXE	RegOpenKey	HKLM\System\CurrentControlSet\Servic...	SUCCESS	Desired Access: AL...
12:50:42	ctfmon.exe	RegOpenKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Desired Access: Q...
12:50:42	Explorer.EXE	RegQueryValue	HKLM\System\CurrentControlSet\Servic...	NAME NOT FOUND	Length: 40
12:50:42	ctfmon.exe	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
12:50:42	Explorer.EXE	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
12:50:42	ctfmon.exe	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
12:50:42	Explorer.EXE	RegOpenKey	HKLM\System\CurrentControlSet\Control...	NAME NOT FOUND	Desired Access: Q...
12:50:42	ctfmon.exe	RegQueryKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Query: Handle Tag...
12:50:42	ctfmon.exe	RegOpenKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Desired Access: Q...
12:50:42	Explorer.EXE	Process Create	C:\Users\Modhupama\Desktop\Ardama...	SUCCESS	PID: 1056, Comman...
12:50:42	ArdamaxKeylogger.exe	Process Create		SUCCESS	Parent PID: 4264, C...
12:50:42	ctfmon.exe	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Type: REG_SZ, Le...
12:50:42	ArdamaxKeylogger.exe	Thread Create		SUCCESS	Thread ID: 2824
12:50:42	ctfmon.exe	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
12:50:42	ctfmon.exe	RegQueryKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Query: Handle Tag...
12:50:42	Explorer.EXE	RegOpenKey	HKCU\Software\Microsoft\Windows\Curr...	SUCCESS	Desired Access: Q...
12:50:42	ctfmon.exe	RegOpenKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Desired Access: Q...
12:50:42	ctfmon.exe	RegQueryValue	HKCU\SOFTWARE\Microsoft\Windows\...	SUCCESS	Type: REG_SZ, Le...
12:50:42	ctfmon.exe	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
12:50:42	ctfmon.exe	RegCloseKey	HKCU\SOFTWARE\Microsoft\Windows\...	SUCCESS	
12:50:42	ctfmon.exe	RegCloseKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
12:50:42	ctfmon.exe	RegOpenKey	HKCU\SOFTWARE\Microsoft\Windows\...	SUCCESS	Desired Access: Q...
12:50:42	ctfmon.exe	RegQueryKey	HKLM	SUCCESS	Query: Handle Tag...
12:50:42	Explorer.EXE	RegQueryValue	HKCU\SOFTWARE\Microsoft\Windows\...	NAME NOT FOUND	Length: 16
12:50:42	ctfmon.exe	RegOpenKey	HKLM\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
12:50:42	Explorer.EXE	RegCloseKey	HKCU\SOFTWARE\Microsoft\Windows\...	SUCCESS	

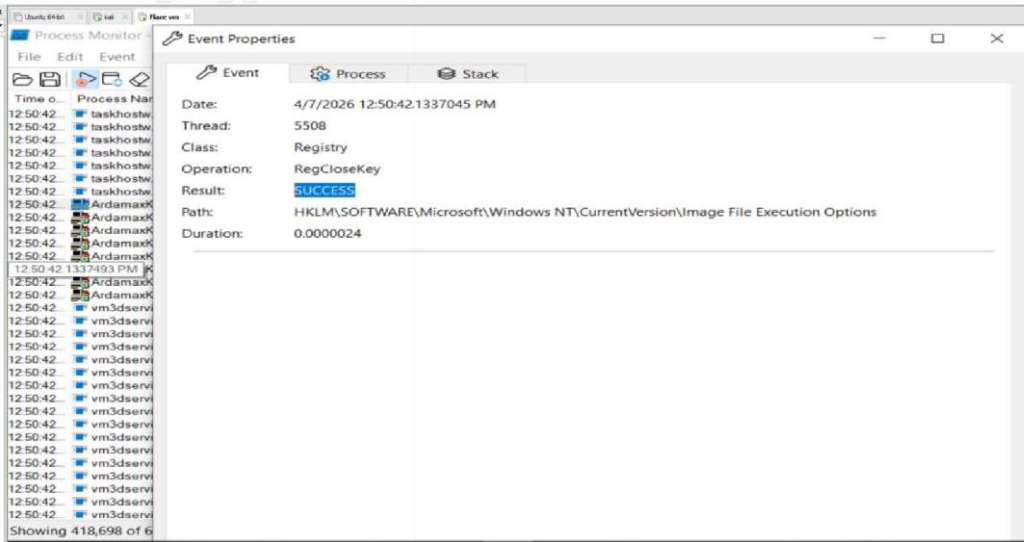
- The malware performs extensive registry enumeration under HKCU\Software\Classes, which may be used to gather system information or identify file associations.
- Repeated RegOpenKey and RegQueryKey operations indicate reconnaissance behavior before performing malicious actions.

Process Monitor - Sysinternals: www.sysinternals.com					
File Edit Event Filter Tools Options Help					
Time	Process Name	Operation	Path	Result	Detail
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Classes\Directory	SUCCESS	Desired Access: R...
12:50:42	ArdamaxKeylogger.exe	RegSetInfoKey	HKCU\Software\Classes\Directory	SUCCESS	KeySetInformation...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes\Directory	SUCCESS	Query: Name
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes\Directory	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Classes\Directory	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Classes\Directory\Shell...	NAME NOT FOUND	Desired Access: Q...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCR\Directory\ShellEx\IconHandler	NAME NOT FOUND	Desired Access: Q...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Classes\Folder	NAME NOT FOUND	Desired Access: R...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCR\Folder	SUCCESS	Desired Access: R...
12:50:42	ArdamaxKeylogger.exe	RegSetInfoKey	HKCR\Folder	SUCCESS	KeySetInformation...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCR\Folder	SUCCESS	Query: Name
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCR\Folder	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Classes\Folder\ShellEx...	NAME NOT FOUND	Desired Access: Q...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCR\Folder	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCR\Folder\ShellEx\IconHandler	NAME NOT FOUND	Desired Access: Q...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Classes\AllFilesystemO...	NAME NOT FOUND	Desired Access: R...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCR\AllFilesystemObjects	SUCCESS	Desired Access: R...
12:50:42	ArdamaxKeylogger.exe	RegSetInfoKey	HKCR\AllFilesystemObjects	SUCCESS	KeySetInformation...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCR\AllFilesystemObjects	SUCCESS	Query: Name
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCR\AllFilesystemObjects	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Classes\AllFilesystemO...	NAME NOT FOUND	Desired Access: Q...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCR\AllFilesystemObjects	SUCCESS	Query: Handle Tag...
12:50:42	ArdamaxKeylogger.exe	RegOpenKey	HKCR\AllFilesystemObjects\ShellEx\Ico...	NAME NOT FOUND	Desired Access: Q...
12:50:42	ArdamaxKeylogger.exe	RegQueryKey	HKCU\Software\Classes\Directory	SUCCESS	Query: Name

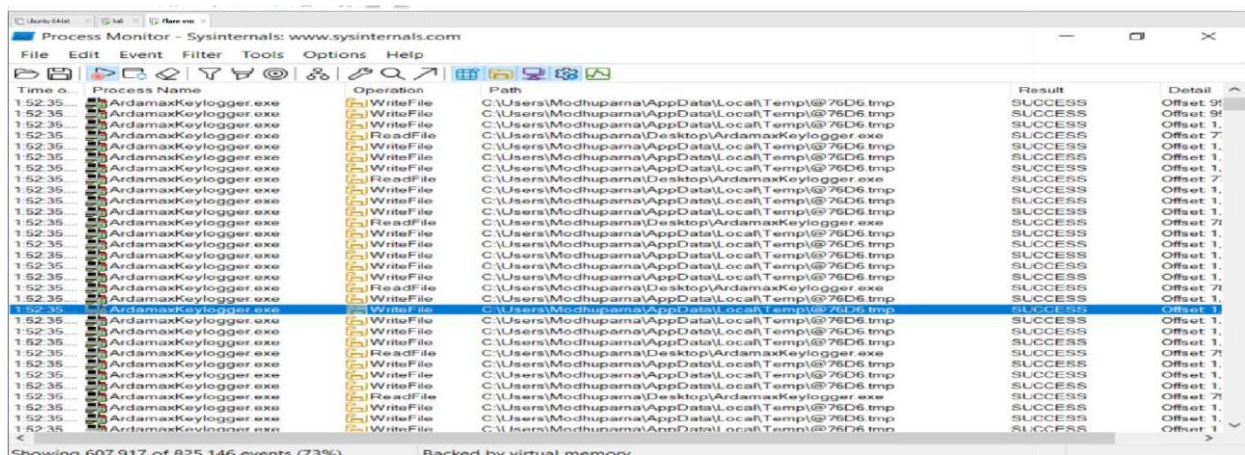
The process accesses

HKLM\SOFTWARE\Microsoft\WindowsNT\CurrentVersion\Image File Execution Options, a registry location commonly abused for persistence and debugging hijacking.

- Interaction with this key suggests the malware may attempt to monitor or modify execution behavior of other programs, which is a known stealth technique.



- This proves:
Malware launched
Executed code
Running internally
- The process ArdamaxKeylogger.exe repeatedly writes data to a temporary file located at:
C:\Users\Modhuparna\AppData\Local\Temp\@76D6.tmp
- Continuous WriteFile operations with fixed buffer size (4096 bytes) indicate structured data logging behavior.
- The use of the Temp directory suggests stealth, as such files are often used to avoid detection and are less monitored.



Interpretation of the File



1. Nature of the File:-

The analyzed .tmp file contains a combination of readable strings, configuration data, and binary content, indicating that it functions as an internal storage or intermediate data file generated by the malware.

- The malware attempts to create/access a file in the system directory (SysWOW64\28463\DPBJ.exe), indicating intent to store or execute data.
- The result NAME NOT FOUND suggests the file does not yet exist, implying an initial file creation attempt.

Time	Process Name	Operation	Path	Result	Detail
1:52:35	ArdamaxKeylogger.exe	RegOpenKey	HKLM\SOFTWARE\Policies\Microsoft...	SUCCESS	Desired Access: Q...
1:52:35	ArdamaxKeylogger.exe	RegSetInfoKey	HKLM\SOFTWARE\Policies\Microsoft...	SUCCESS	KeySetInformation...
1:52:35	ArdamaxKeylogger.exe	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft...	NAME NOT FOUND	Length: 16
1:52:35	ArdamaxKeylogger.exe	RegCloseKey	HKLM\SOFTWARE\Policies\Microsoft...	SUCCESS	
1:52:35	ArdamaxKeylogger.exe	RegQueryKey	HKCU	SUCCESS	Query: Handle Tag...
1:52:35	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Policies\MicrosoftWind...	NAME NOT FOUND	Query: Name
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Windows\SysWOW64\28463	REPARSE	Desired Access: R...
1:52:35	ArdamaxKeylogger.exe	QueryRemoteP...	C:\Users\Modhupama\AppData\Local\V...	INVALID PARAM...	Desired Access: R...
1:52:35	ArdamaxKeylogger.exe	QueryDirectory	C:\Users\Modhupama\AppData\Local\V...	NO SUCH FILE	FileInformationClas...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\windows.stora...	SUCCESS	Offset 951,296. Len...
1:52:35	ArdamaxKeylogger.exe	CloseFile	C:\Users\Modhupama\AppData\Local\V...	SUCCESS	
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Windows\SysWOW64\28463\DPBJ.e...	NAME NOT FOUND	Desired Access: R...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\shell32.dll	SUCCESS	Offset 963,584. Len...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\shell32.dll	SUCCESS	Offset 91,136. Len...
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Users\Modhupama\Desktop\PROPS...	NAME NOT FOUND	Desired Access: R...
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Desired Access: R...
1:52:35	ArdamaxKeylogger.exe	QueryBasicInfor...	C:\Windows\SysWOW64\propsys.dll	SUCCESS	CreationTime: 2/12/...
1:52:35	ArdamaxKeylogger.exe	CloseFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Desired Access: R...
1:52:35	ArdamaxKeylogger.exe	CreateFileMap...	C:\Windows\SysWOW64\propsys.dll	FILE LOCKED WIT...	SyncType: SyncTy...
1:52:35	ArdamaxKeylogger.exe	QueryStandard...	C:\Windows\SysWOW64\propsys.dll	SUCCESS	AllocationSize: 790...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 0. Length: 4.0...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 37,040. Leng...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 13,312. Leng...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 54,272. Leng...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 728,576. Len...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 744,960. Len...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 761,344. Len...
1:52:35	ArdamaxKeylogger.exe	CreateFileMap...	C:\Windows\SysWOW64\propsys.dll	SUCCESS	SyncType: SyncTy...
1:52:35	ArdamaxKeylogger.exe	Load Image...	C:\Windows\SysWOW64\propsys.dll	SUCCESS	ImageBase: 0x74B...

- The malware requests read, write, and delete permissions, confirming its capability to modify or create files.
- Such access rights strongly indicate data storage behavior, commonly seen in keyloggers.

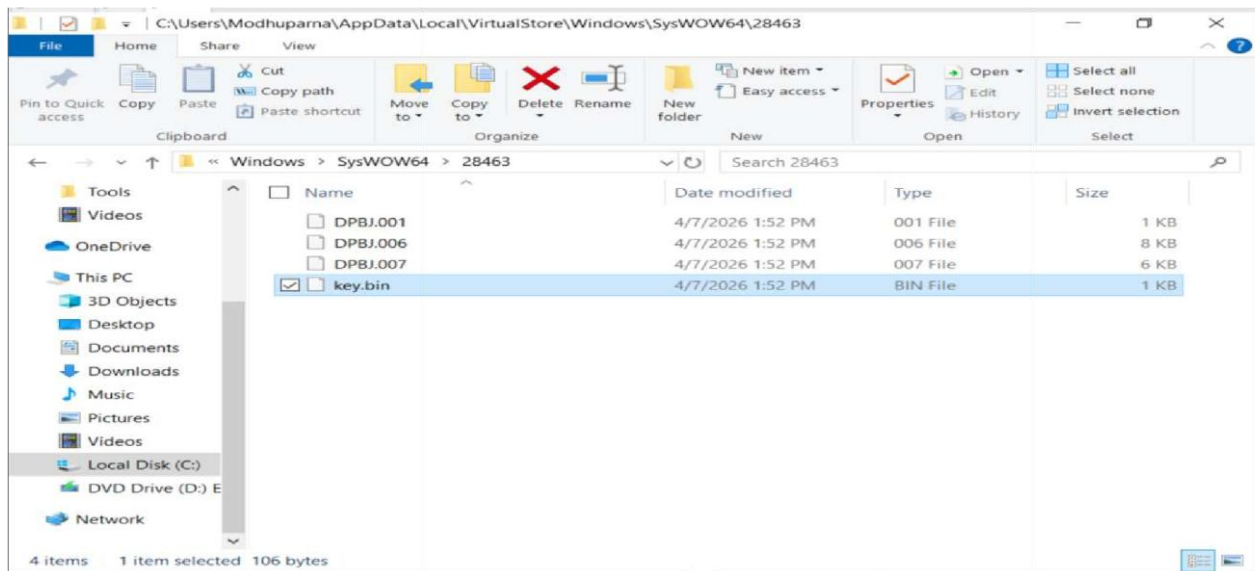
Event	Process	Stack
Date:	4/7/2026 1:52:35.6086338 PM	
Thread:	5456	
Class:	File System	
Operation:	CreateFile	
Result:	NAME NOT FOUND	
Path:	C:\Windows\SysWOW64\28463\DPBJ.exe	
Duration:	0.0000122	
Desired Access:	Read Attributes, Synchronize	
Disposition:	Open	
Options:	Synchronous IO Non-Alert	
Attributes:	N	
ShareMode:	Read, Write, Delete	
AllocationSize:	n/a	

- The file operation is redirected to AppData\Local\VirtualStore, showing the malware attempted to write in a protected system location.

- This confirms that the malware is actively trying to persist data on the system despite permission restrictions.

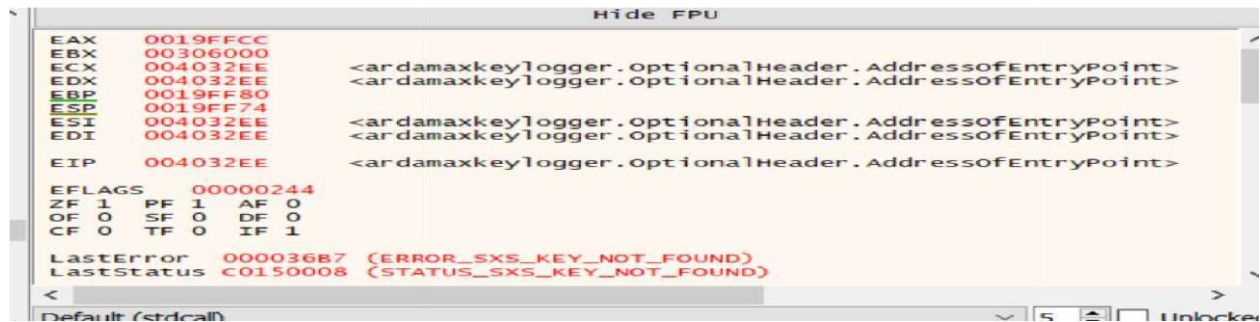
Time	Process Name	Operation	Path	Result	Detail
1:52:35	ArdamaxKeylogger.exe	RegOpenKey	HKLM\SOFTWARE\Policies\Microsoft\Windows\Explorer	SUCCESS	Desired
1:52:35	ArdamaxKeylogger.exe	RegSetInfoKey	HKLM\SOFTWARE\Policies\Microsoft\Windows\Explorer	SUCCESS	KeySettr
1:52:35	ArdamaxKeylogger.exe	RegQueryVal...	HKLM\SOFTWARE\Policies\Microsoft\Windows\Explorer\UseFindFirstF...	NAME NOT FOUND	Length: 1
1:52:35	ArdamaxKeylogger.exe	RegCloseKey	HKLM\SOFTWARE\Policies\Microsoft\Windows\Explorer	SUCCESS	
1:52:35	ArdamaxKeylogger.exe	RegQueryKey	HKCU	SUCCESS	Query N...
1:52:35	ArdamaxKeylogger.exe	RegOpenKey	HKCU\Software\Policies\Microsoft\Windows\Explorer	NAME NOT FOUND	Desired
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Windows\SysWOW64\28463	REPARSE	Desired
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Users\Modhuparna\AppData\Local\VirtualStore\Windows\SysWOW...	SUCCESS	Desired
1:52:35	ArdamaxKeylogger.exe	QueryRemote...	C:\Users\Modhuparna\AppData\Local\VirtualStore\Windows\SysWOW...	INVALID PARAME...	
1:52:35	ArdamaxKeylogger.exe	QueryDirectory	C:\Users\Modhuparna\AppData\Local\VirtualStore\Windows\SysWOW...	NO SUCH FILE	FileInfor...
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\windows.storage.dll	SUCCESS	Offset 9t
1:52:35	ArdamaxKeylogger.exe	CloseFile	C:\Users\Modhuparna\AppData\Local\VirtualStore\Windows\SysWOW...	SUCCESS	
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Windows\SysWOW64\28463\DPBJ.exe	NAME NOT FOUND	Desired
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\shell32.dll	SUCCESS	Offset 9t
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\shell32.dll	SUCCESS	Offset 9'
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Users\Modhuparna\Desktop\PROPSYS.dll	NAME NOT FOUND	Desired
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Desired
1:52:35	ArdamaxKeylogger.exe	QueryBasicInf...	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Creation
1:52:35	ArdamaxKeylogger.exe	CloseFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	
1:52:35	ArdamaxKeylogger.exe	CreateFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Desired
1:52:35	ArdamaxKeylogger.exe	CreateFileMa...	C:\Windows\SysWOW64\propsys.dll	FILE LOCKED WIT...	SyncTyr
1:52:35	ArdamaxKeylogger.exe	QueryStandar...	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Allocatic
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 0.
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 8:
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 1:
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 5:
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 7:
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 7:
1:52:35	ArdamaxKeylogger.exe	ReadFile	C:\Windows\SysWOW64\propsys.dll	SUCCESS	Offset 7t
1:52:35	ArdamaxKeylogger.exe	CreateFileMa...	C:\Windows\SysWOW64\propsys.dll	SUCCESS	SyncTyr

- The presence of files like key.bin and DPBJ.* confirms that the malware successfully created files to store data.
- The key.bin file strongly indicates storage of captured keystrokes, validating keylogging behavior.



x32 dbg

Entry point



Hide FPU

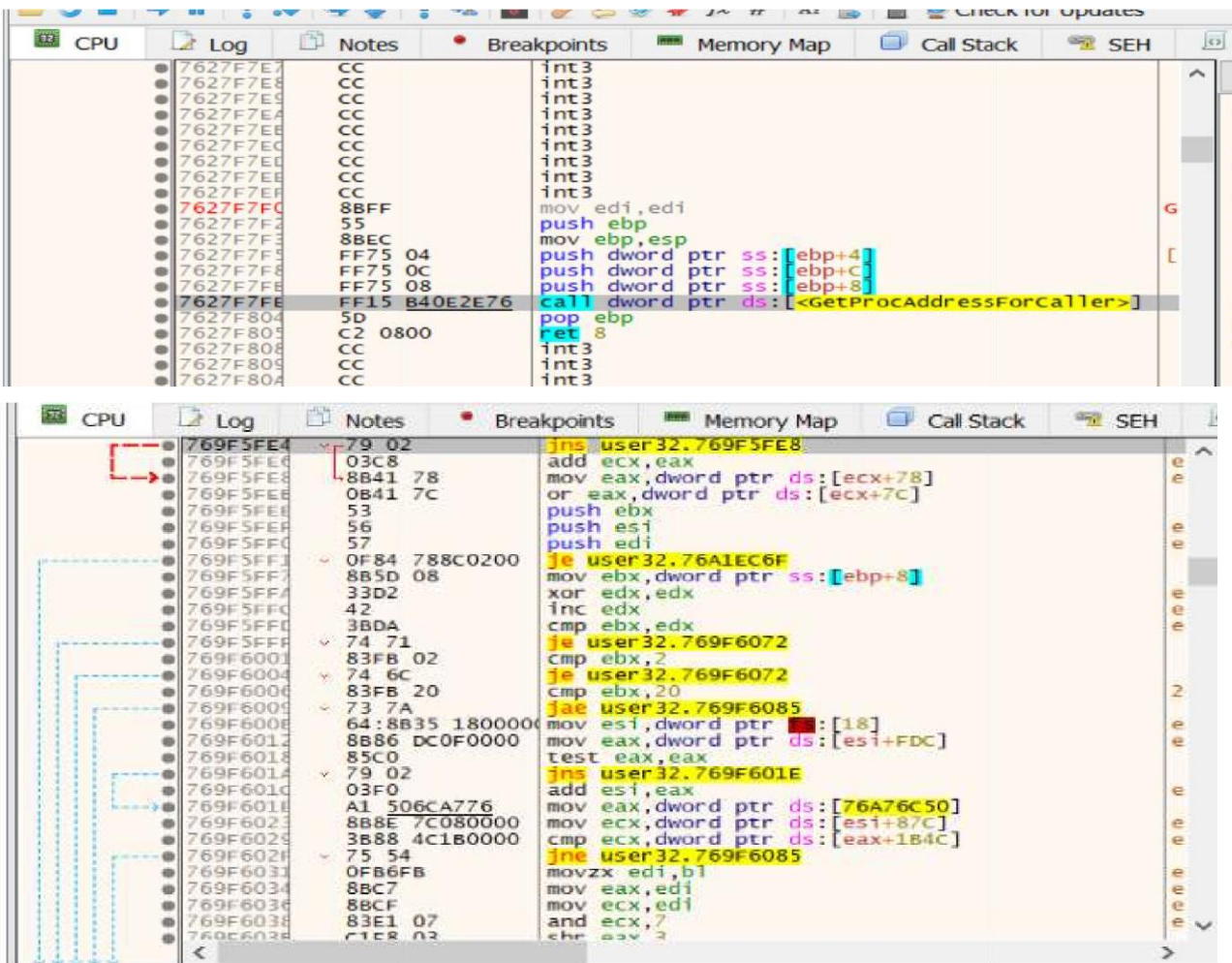
EAX	0019FFCC	
EBX	00306000	<ardamaxkeylogger.OptionalHeader.AddressOfEntryPoint>
ECX	004032EE	<ardamaxkeylogger.OptionalHeader.AddressOfEntryPoint>
EDX	004032EE	<ardamaxkeylogger.OptionalHeader.AddressOfEntryPoint>
EBP	0019FF80	
ESP	0019FF74	
ESI	004032EE	<ardamaxkeylogger.OptionalHeader.AddressOfEntryPoint>
EDI	004032EE	<ardamaxkeylogger.OptionalHeader.AddressOfEntryPoint>
EIP	004032EE	<ardamaxkeylogger.OptionalHeader.AddressOfEntryPoint>

EFLAGS 00000244
ZF 1 PF 1 AF 0
OF 0 SF 0 DF 0
CF 0 TF 0 IF 1

LastError 000036B7 (ERROR_SXS_KEY_NOT_FOUND)
LastStatus C0150008 (STATUS_SXS_KEY_NOT_FOUND)

Default (stdcall)

- The malware calls GetProcAddress, indicating dynamic API resolution instead of static imports.
- This technique is used to hide malicious functions (like keylogging APIs) and evade detection.

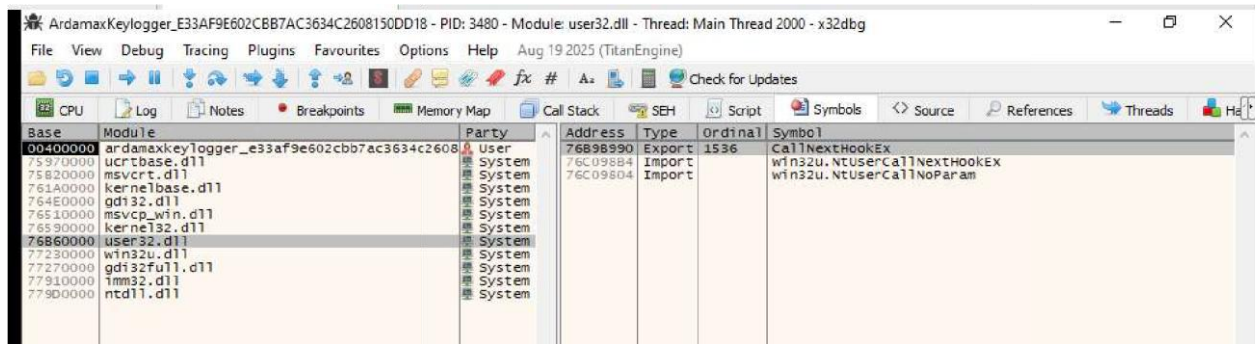


CPU Log Notes Breakpoints Memory Map Call Stack SEH

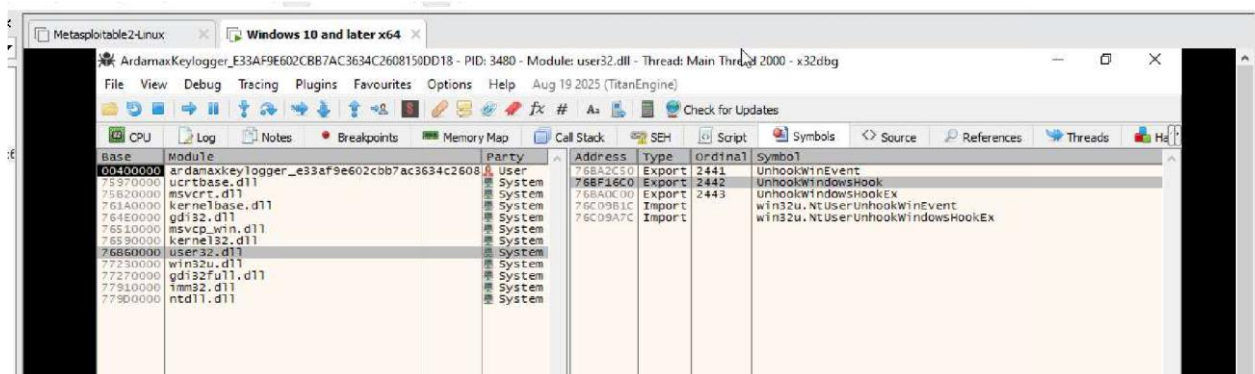
7627F7E7 CC int3
7627F7E8 CC int3
7627F7E9 CC int3
7627F7EA CC int3
7627F7EB CC int3
7627F7EC CC int3
7627F7ED CC int3
7627F7EE CC int3
7627F7EF CC int3
7627F7F0 8BFF mov edi,edi
7627F7F1 55 push ebp
7627F7F2 8BEC mov ebp,esp
7627F7F3 FF75 04 push dword ptr ss:[ebp+4]
7627F7F4 FF75 0C push dword ptr ss:[ebp+C]
7627F7F5 FF75 08 push dword ptr ss:[ebp+8]
7627F7F6 FF15 B40E2E76 call dword ptr ds:[<GetProcAddressForCaller>]
7627F804 5D pop ebp
7627F805 C2 0800 ret 8
7627F806 CC int3
7627F807 CC int3
7627F808 CC int3
7627F809 CC int3
7627F80A CC int3

CPU Log Notes Breakpoints Memory Map Call Stack SEH

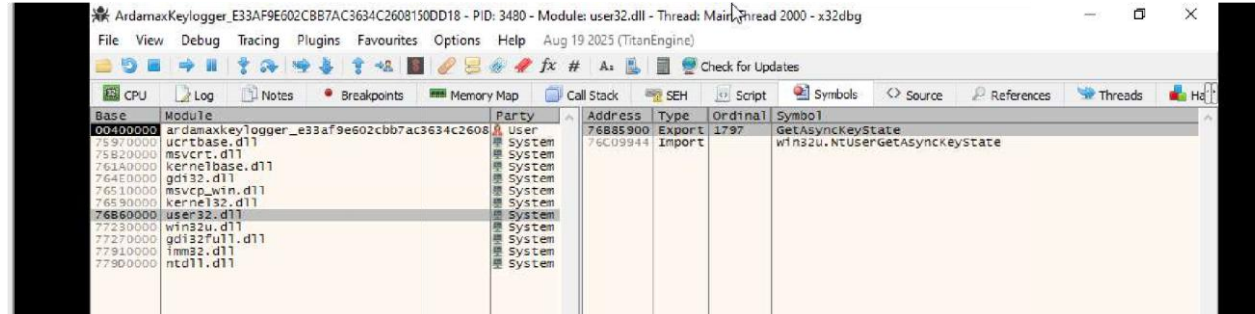
769F5FE4 79 02 jns user32.769F5FE8
769F5FE5 03C8 add ecx,eax
769F5FE6 8B41 78 mov eax,dword ptr ds:[ecx+78]
769F5FE7 0B41 7C or eax,dword ptr ds:[ecx+7C]
769F5FE8 53 push ebx
769F5FE9 56 push esi
769F5FEA 57 push edi
769F5FEB 0F84 788C0200 je user32.76A1EC6F
769F5FEC 8B5D 08 mov ebx,dword ptr ss:[ebp+8]
769F5FED 33D2 xor edx,edx
769F5FEE 42 inc edx
769F5FEF 3BD0 cmp ebx,edx
769F5FF0 74 71 je user32.769F6072
769F6001 83FB 02 cmp ebx,2
769F6002 74 6C je user32.769F6072
769F6003 83FB 20 cmp ebx,20
769F6004 73 7A jae user32.769F6085
769F6005 64:8B35 18000000 mov esi,dword ptr [18]
769F6006 8B86 DC0F0000 mov eax,dword ptr ds:[esi+FDC]
769F6007 85C0 test eax,eax
769F6008 79 02 jns user32.769F601E
769F6009 03F0 add esi,eax
769F600A A1 506CA776 mov eax,dword ptr ds:[76A76C50]
769F600B 8B8E 7C080000 mov ecx,dword ptr ds:[esi+87C]
769F600C 3B88 4C1B0000 cmp ecx,dword ptr ds:[eax+1B4C]
769F600D 75 54 jne user32.769F6085
769F600E 0FB6FB movzx edi,bl
769F600F 8BC7 mov eax,edi
769F6010 8BCF mov ecx,edi
769F6011 83E1 07 and ecx,7
769F6012 C1E8 03 shr eax,3



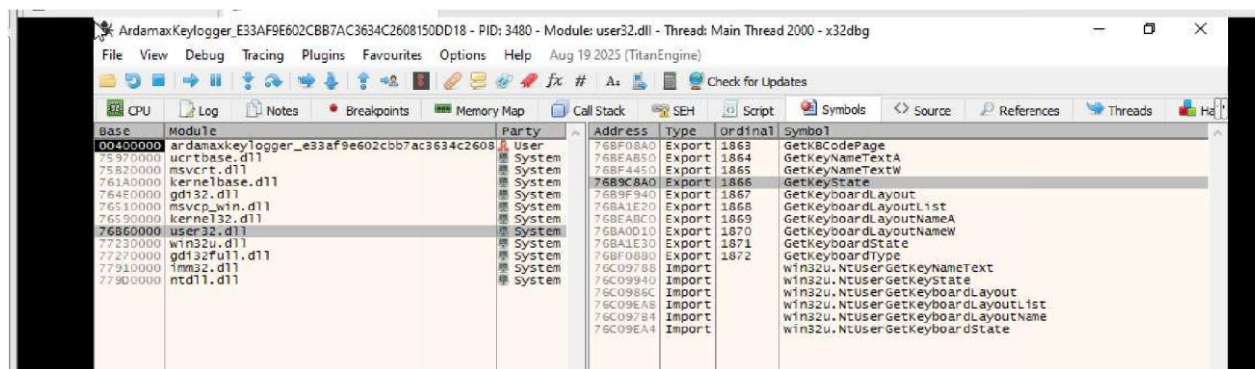
UnHookWindowsHook

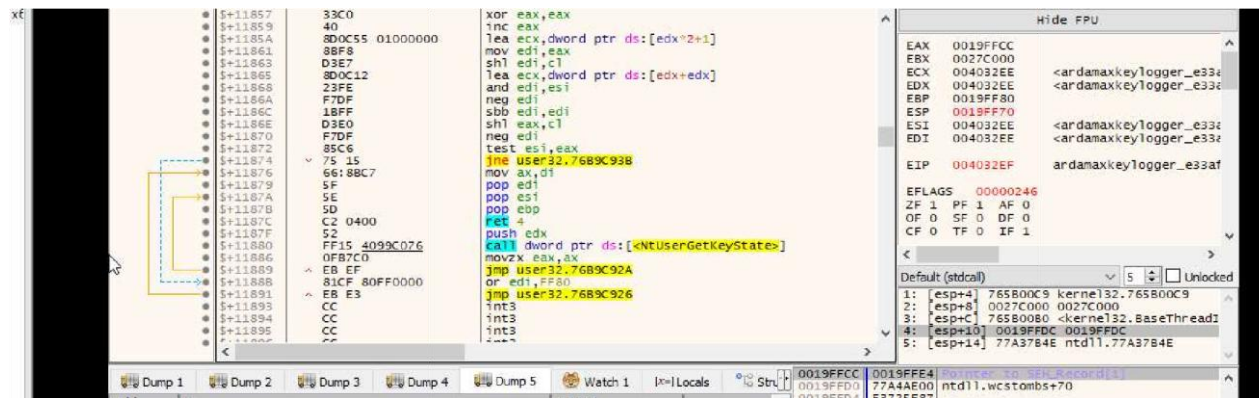


GetAsyncKeyState

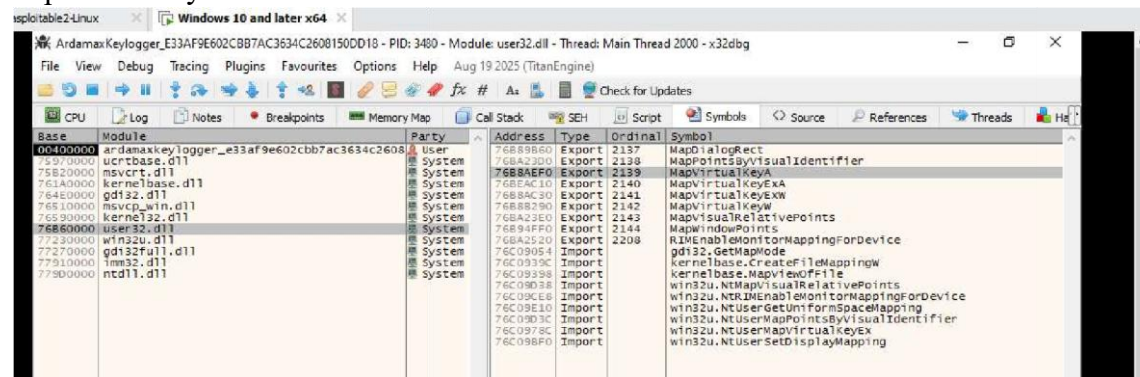


GetKeyState

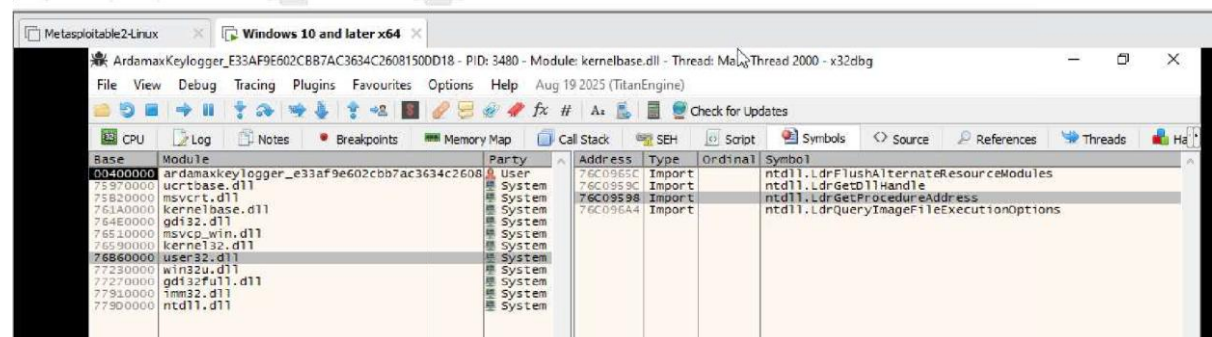
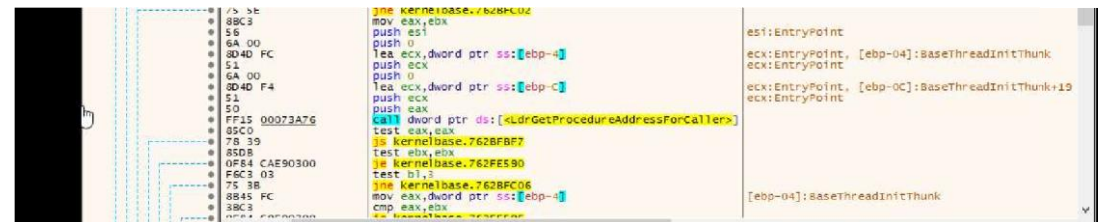




MapVirtualKeyA



GetProcAddress



LdrGetProcedureAddress

During execution, no significant outbound or inbound network traffic related to data transmission was observed. The absence of such activity is expected due to the restricted network configuration of the analysis environment.

However, it is important to note that the lack of observable network communication does not necessarily imply that the malware lacks communication capabilities. Under normal conditions, the malware may attempt to establish connections with external Command and Control (C2) servers.

Additionally, the sample may incorporate techniques such as delayed execution, environmental awareness, or sandbox evasion, which can suppress network activity when executed in isolated or analysis environments.

Further investigation using controlled network simulation tools such as FakeNet and Wireshark revealed only limited network activity, including ICMP echo requests, NBNS queries, and ARP communication. These activities are indicative of basic network probing and local network interactions rather than active data transmission.

No evidence of data exfiltration, SMTP-based communication, or interaction with external Command and Control infrastructure was identified during the analysis.

Based on the observed behavior, the analyzed sample appears to primarily focus on local data collection mechanisms, such as keystroke logging and file storage, without initiating network-based data transfer in the given environment.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.209.128	8.8.8.8	ICMP	98	Echo (ping) request id
2	2.013954	192.168.209.128	8.8.4.4	ICMP	98	Echo (ping) request id
3	4.030909	192.168.209.128	1.1.1.1	ICMP	98	Echo (ping) request id
4	6.030986	192.168.209.128	1.0.0.1	ICMP	98	Echo (ping) request id
5	7.386279	192.168.209.128	192.168.209.255	NBNS	92	Name query NB DESKTOP-Q
6	7.386833	192.168.209.128	192.168.209.255	NBNS	92	Name query NB DESKTOP-Q
7	8.143900	192.168.209.128	192.168.209.255	NBNS	92	Name query NB DESKTOP-Q
8	8.146838	192.168.209.128	192.168.209.255	NBNS	92	Name query NB DESKTOP-Q
9	8.906176	192.168.209.128	192.168.209.255	NBNS	92	Name query NB DESKTOP-Q
10	8.907593	192.168.209.128	192.168.209.255	NBNS	92	Name query NB DESKTOP-Q
11	10.063150	192.168.209.128	8.8.8.8	ICMP	98	Echo (ping) request id
12	12.080837	192.168.209.128	8.8.4.4	ICMP	98	Echo (ping) request id
13	14.077871	192.168.209.128	1.1.1.1	ICMP	98	Echo (ping) request id
14	16.094019	192.168.209.128	1.0.0.1	ICMP	98	Echo (ping) request id
15	20.124647	192.168.209.128	8.8.8.8	ICMP	98	Echo (ping) request id
16	22.126475	192.168.209.128	8.8.4.4	ICMP	98	Echo (ping) request id
17	24.126898	192.168.209.128	1.1.1.1	ICMP	98	Echo (ping) request id
18	26.138467	192.168.209.128	1.0.0.1	ICMP	98	Echo (ping) request id
19	26.638060	VMware_c4:f6:57	VMware_c0:00:01	ARP	42	Who has 192.168.209.1?
20	26.638197	VMware_c0:00:01	VMware_c4:f6:57	ARP	60	192.168.209.1 is at 00:
21	30.154947	192.168.209.128	8.8.8.8	ICMP	98	Echo (ping) request id

5.Indicators of Compromise (IOCs):

The following indicators were identified during analysis:

- File Name: ArdamaxKeylogger.exe File Type: PE32 Executable
- Temporary File Created:

- C:\Users\Modhuparna\AppData\Local\Temp\@76D6.tmp
- Additional Files Created:
 - key.bin ◦ DPBJ.exe ◦ DPBJ.*
- Registry Keys Accessed:
 - HKLM\SOFTWARE\Microsoft\Input\Settings
 - HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options
- Suspicious API Calls: ◦ SetWindowsHookEx ◦ GetAsyncKeyState ◦ CallNextHookEx ◦ MapVirtualKey
- Extracted Strings Indicators:
 - Keystrokes Log ◦ Web Log ◦ Chat Log ◦ Screenshots ◦ SMTP Host / Port ◦ FTP

7. Attack Chain Analysis

The behavior of the malware can be summarized in the following attack chain:

1. Execution:

The user executes the malicious file (ArdamaxKeylogger.exe), initiating the malware process.
2. Initialization & Evasion:

The malware uses packed/obfuscated code and dynamic API resolution to evade detection.
3. System Interaction:

It accesses registry keys and system APIs related to input handling and file operations.
4. Keylogging Setup:

A keyboard hook is installed using SetWindowsHookEx, allowing interception of keystrokes.
5. Data Collection:

The malware captures:

 - Keystrokes ◦
 - Web activity ◦
 - Chat logs ◦
 - Screenshots
6. Local Storage:

Captured data is written into temporary files (e.g., .tmp, key.bin) in structured format.
7. Data Preparation:

Data may be encoded or formatted for transmission.
8. Exfiltration (Potential):

The malware is designed to send data via SMTP or FTP (not observed due to isolation).

9. Persistence/Stealth:

Attempts to interact with system directories and registry for continued operation.

8. Key Findings:-

- The analyzed sample is a keylogger spyware capable of monitoring multiple user activities.
- Static analysis confirmed packed/obfuscated structure, indicating anti-analysis techniques. □
Dynamic analysis revealed continuous file writing in the Temp directory, confirming active data logging.
- Extracted strings showed structured logging modules such as keystrokes, web logs, chat logs, and screenshots.
- The presence of SMTP and FTP strings indicates data exfiltration capability.
- Keyboard hook APIs confirm low-level interception of user input.
- The key.bin file contained actual captured user input, providing direct proof of keylogging. □
No network exfiltration was observed due to controlled offline environment, not lack of capability.

9. Conclusion:

The analyzed sample, ArdamaxKeylogger.exe, was confirmed to be a keylogger and spywaretype malware. Static analysis indicated that the file is a packed PE32 executable utilizing Windows API functions related to file handling and user input monitoring. Dynamic analysis further confirmed malicious behavior, including registry access, thread creation, and continuous writing of captured data into temporary files.

The presence of structured logging indicators such as keystroke logs, web logs, chat logs, and screenshot capture demonstrates that the malware performs comprehensive user activity monitoring. Additionally, extracted strings revealed SMTP and FTP configuration parameters, indicating the capability for data exfiltration.

Although no network-based data transfer was observed due to the isolated analysis environment, the malware clearly exhibits behavior consistent with credential theft, surveillance, and unauthorized data collection. The presence of actual captured input in the key.bin file further confirms active keylogging functionality.

Overall, the malware poses a significant threat to user privacy and system security, emphasizing the importance of proactive detection and secure computing practices.